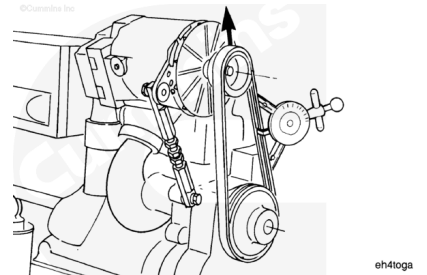


Trainings ▾

Test

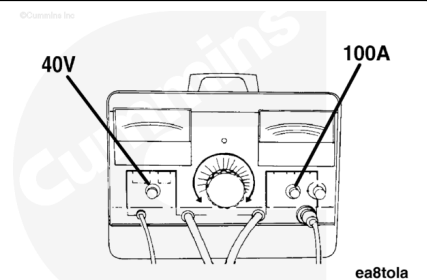
Before performing the following test, be sure the alternator belt is tightened to the correct specifications. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/56/56-013-005.html>).

The following instructions are for use with the charging and cranking system analyzer, Part Number 3885847, or equivalent.



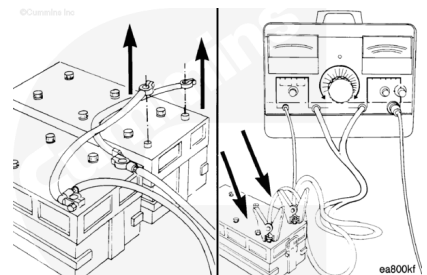
Set the voltage selector knob to the appropriate scale. For a 24-VDC system, choose the 40-VDC scale.

Set the amp selector knob to 100 amps.



⚠ WARNING ⚠

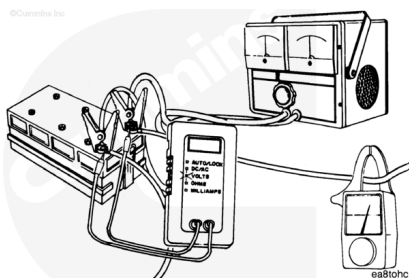
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Remove the cables to any other battery in the circuit.

Connect the correct analyzer leads to the positive (+) and negative (-) terminals on the battery.

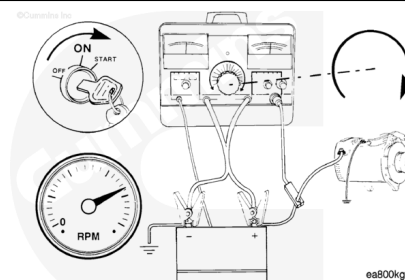
Connect the clamp-on amp pickup to the alternator output cable as far away from the alternator as possible.



Operate the engine at high idle, and turn the analyzer load control knob **clockwise** until a maximum amp reading is obtained.

Do **not** let the load volts drop below 26-VDC for a 24-VDC system.

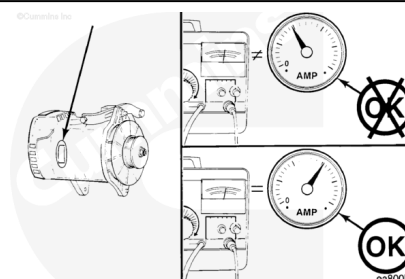
The maximum amp reading is the alternator output, and **must** meet the alternator manufacturer's specifications.



The alternator maximum-rated output is normally stamped or labeled on the alternator.

Also, check the equipment ammeter gauge. If it does **not** read approximately the same as the test equipment, it **must** be replaced.

If the alternator output is **not** within 10 percent of rated output, repair or replace the alternator. Refer to the manufacturer's instructions.

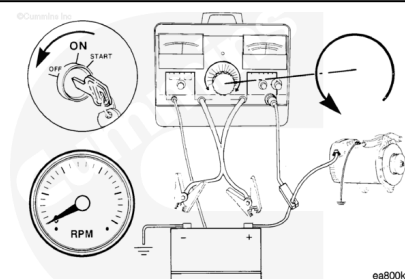


⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Turn the analyzer load control knob **counterclockwise** to the OFF position, and shut off the engine.

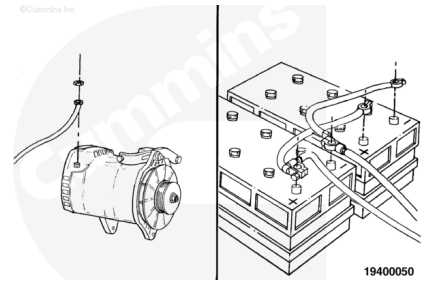
Remove the test equipment. Connect all battery cables that were removed.



Remove

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Disconnect the batteries.

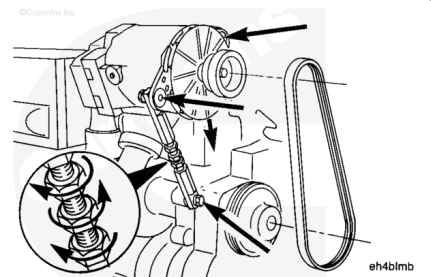
Disconnect the wiring from the alternator.

Loosen the adjusting link and the alternator mounting capscrews.

The lower jam nut has left-hand threads.

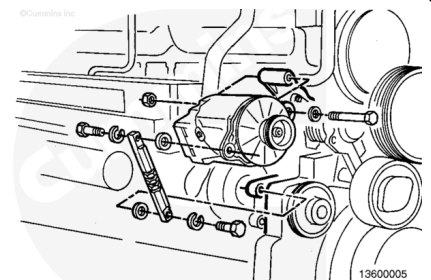
Loosen both of the jam nuts. Turn the adjusting screw to relieve the belt tension.

Remove the alternator belt.



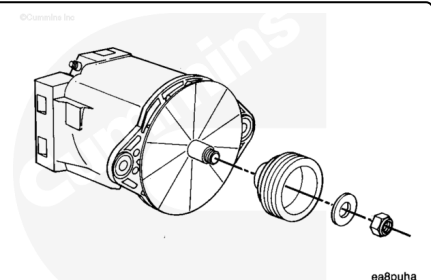
The three capscrews are SAE Grade 8 and the three spacers are hardened. Keep the parts together to aid the assembly procedure.

Remove the parts as shown.



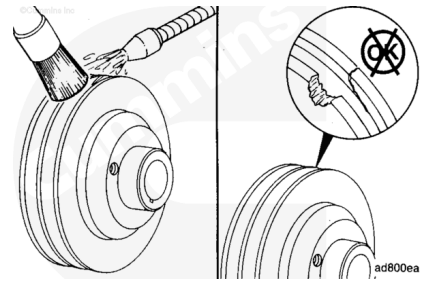
Clean and Inspect for Reuse

Remove the nut and pulley from the alternator.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying dirt and debris can cause personal injury.

Use solvent or a steam cleaner to clean the pulley and dry with compressed air.

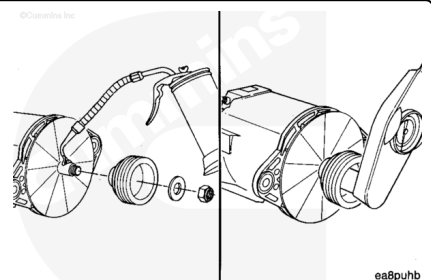
Check the grooves of the pulley for wear and damage.

If worn or damaged, replace the pulley.

Install

Lubricate the shaft with clean engine oil. Install the pulley and nut on the alternator shaft. Tighten the nut.

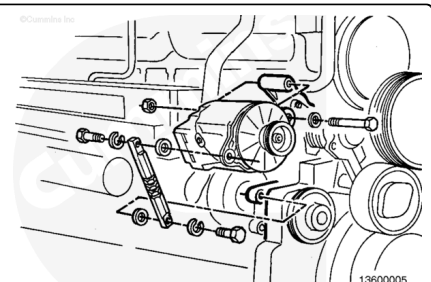
Torque Value: 95 N·m [70 ft·lb]



The belt **must** be installed and adjusted before the capscrews are tightened.

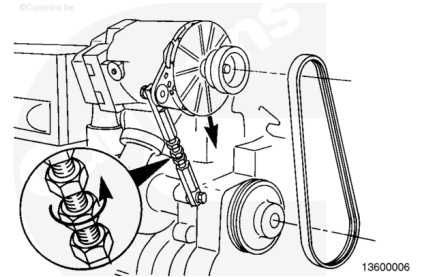
The end of the adjusting link with the largest area at the cap screw hole **must** be nearest to the alternator.

Install the alternator and the adjusting link, as shown.



Do **not** attempt to pry the belt on the pulley.

Install the belt. Turn the adjusting screw **counterclockwise** to shorten the link if necessary.



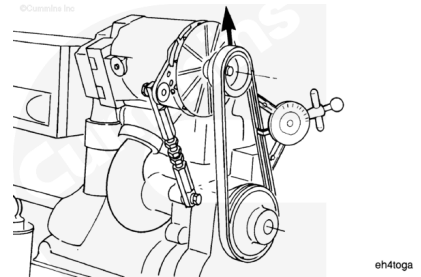
Use a belt tension gauge (Borroughs), Part Number ST-1293, to measure the poly V-belt tension.

Refer to the Drive Belt Tension Table in Section V.

A belt is considered used if it has been in operation for 10 minutes or longer.

If used belt tension is below the minimum value, tighten to the maximum value.

Replace the belt if it will **not** maintain the correct tension.



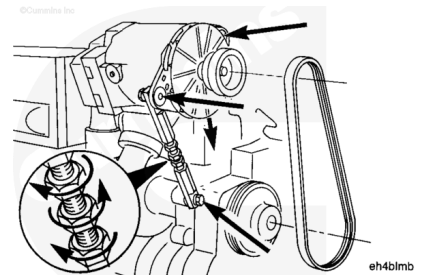
The lower jam nut has left-hand threads.

Tighten the jam nuts on the adjusting screw.

Torque Value: 80 n•m [59 ft-lb]

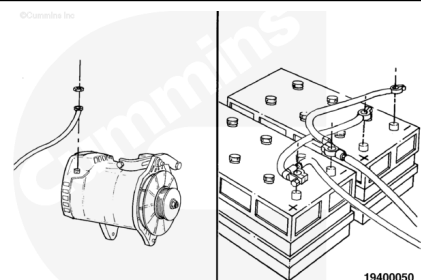
Tighten the adjusting link and alternator mounting capscrews.

Torque Value: 115 n•m [85 ft-lb]



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Connect the wiring to the alternator.

Connect the batteries.

